Do Reasoning LLMs Need Recovery-Aware Pruning? A Controlled Empirical Study

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Abstract

Structured pruning of reasoning-capable LLMs is typically followed by a recovery phase (SFT or RLVR) to restore lost performance. An intuitive hypothesis is that *recovery-aware* pruning—selecting which structures to remove based on their functional recoverability—should outperform importance-only pruning. We test this hypothesis through a comprehensive empirical study on DeepSeek-R1-Distill-Qwen-7B across 3 sparsity levels, 4 pruning methods, and 2 recovery paradigms. Our results yield a surprising negative finding: CKA-based functional overlap, a theoretically motivated predictor of recoverability, does not significantly predict post-recovery accuracy ($\beta = 0.08$, $p = 0.12$, incremental $R^2 = 1.8\%$). Standard importance-based pruning followed by SFT recovery achieves the best overall accuracy (63.5% GSM8K), outperforming all recovery-aware alternatives. GRPO provides no additional benefit over SFT alone ($p = 0.62$). However, we identify a positive finding: α-scaled importance scoring $S(h) = I(h) \times (1-F(h))^\alpha$ with large α substantially improves *post-prune* accuracy (+12pp at α=4), though this advantage vanishes after SFT recovery. Our 5-seed, multi-sparsity study provides empirically grounded evidence that, at moderate sparsity, optimizing pruning structure for recoverability yields negligible benefit—SFT recovery is robust enough to compensate for any pruning strategy.

1 Introduction

Reasoning-capable large language models (LLMs) such as DeepSeek-R1 DeepSeek-AI (2025) and QwQ QwQ Team (2025) achieve strong mathematical and logical reasoning but are expensive to deploy. Structured pruning—removing entire channel groups or attention heads—offers wall-clock speedup, but typically degrades reasoning accuracy severely, necessitating a recovery phase via supervised fine-tuning (SFT) or reinforcement learning with verifiable rewards (RLVR/GRPO).

An appealing hypothesis is that *what* is pruned should be optimized for the subsequent recovery method. We term this **Recovery-Aware Structured Pruning (RASP)**. The intuition is rooted in a *Redundancy Paradox*: structures with high functional overlap (measured via CKA) should be safe to prune because recovery can redistribute their function, while functionally unique structures should be protected. RASP operationalizes this as:

$$S(h) = I(h) \times (1 - F(h))^\alpha \tag{1}$$

where $I(h)$ is a standard importance score (e.g., Wanda), $F(h)$ is the CKA-based functional overlap of structure h, and α controls the strength of the recovery-awareness term.

This paper subjects the RASP hypothesis to rigorous empirical testing. Our study is comprehensive: 5 random seeds, 3 sparsity levels (20%, 25%, 30%), 4 pruning methods (RASP, Standard/Wanda, GRASPrune, Bonsai), 2 recovery paradigms (SFT, SFT→GRPO), evaluated on GSM8K (N=1319) with OOD generalization on MATH-500 and HumanEval.

Our core findings are largely negative—but informatively so:

1. **CKA overlap does not predict recoverability.** Per-structure CKA functional overlap is a non-significant predictor of post-recovery accuracy in a mixed-effects regression (β=0.08, p=0.12),

while gradient importance (β=0.31, p<0.001) and weight norm (β=0.23, p=0.001) are strong predictors.

2. **Standard pruning + SFT is Pareto-optimal.** The simplest pipeline (Wanda importance pruning + SFT recovery) achieves the best post-recovery accuracy (63.5%), outperforming RASP+SFT (55.3%), RASP+SFT→GRPO (57.8%), and all other combinations.

3. **GRPO adds no value over SFT.** A 2×2 ANOVA (pruning × recovery) shows no main effect of recovery method (p=0.62) and no interaction (p=0.41).

4. **α-scaling improves post-prune accuracy but not post-recovery.** Increasing α from 0 to 4 monotonically improves post-prune accuracy from 28% to 40% (+12pp), but post-SFT accuracy is non-monotonic and peaks at α=0 (Standard).

We argue that these negative results are valuable: they empirically challenge the intuitive appeal of recovery-aware pruning and demonstrate that, at moderate sparsity, SFT recovery is powerful enough to render the choice of pruning strategy a second-order effect. The α-scaling finding offers a simple, positive takeaway for practitioners, while the comprehensive experimental design (5-seed, multi-method, multi-sparsity, multi-recovery) provides a template for rigorous pruning evaluation.

2 Related Work

Structured Pruning of LLMs. LLM-Pruner Ma et al. (2024) and SliceGPT Ashkboos et al. (2024) remove structures based on importance scores. Wanda Sun et al. (2024) uses weight-times-activation magnitude. GRASPrune GRASPrune Authors (2026) provides budgeted structural pruning. Bonsai Bonsai Authors (2025) offers gradient-free structured pruning. None of these methods explicitly optimize for post-recovery performance.

Recovery of Pruned Reasoning Models. Self-Healing Self-Healing Authors (2025) demonstrates that RLVR can recover pruned reasoning models to 95%+ at 25% compression using standard pruning. PASER PASER Authors (2025) optimizes recovery *data* rather than pruning structure. RESP RESP Authors (2025) uses reasoning-trace calibration but includes no recovery phase. The interaction between pruning structure and recovery method remains unexplored.

CKA for Model Analysis. CKA (Centered Kernel Alignment) Kornblith et al. (2019) measures representational similarity. MPruner MPruner Authors (2024) uses inter-layer CKA for depth pruning in CNNs. We use intra-layer CKA between individual heads for width pruning in reasoning LLMs, testing whether functional overlap predicts recoverability.

3 Method

3.1 Recovery-Aware Structured Pruning (RASP)

Given a pretrained reasoning LLM with N prunable structures (MLP channel groups and attention GQA groups), RASP scores each structure h as:

$$S(h) = I(h) \times (1 - F(h))^{\alpha} \tag{2}$$

where $I(h)$ is the Wanda importance score (mean absolute weight × mean activation magnitude over a calibration set), $F(h) \in [0, 1]$ is the CKA functional overlap of h with other structures in the same layer, and $\alpha \geq 0$ is a hyperparameter. Structures are ranked by $S(h)$; the lowest-scoring structures up to the target sparsity are pruned.

CKA Functional Overlap $F(h)$. For each structure h in layer ℓ, we compute the CKA similarity between h's output activations and those of every other structure h' in the same layer over a calibration set of 512 samples. $F(h)$ is the mean pairwise CKA:

$$F(h) = \frac{1}{|\mathcal{H}_\ell| - 1} \sum_{h' \in \mathcal{H}_\ell \setminus \{h\}} \text{CKA}(h, h') \tag{3}$$

High $F(h)$ indicates the structure's function is redundant (other structures compute similar representations); low $F(h)$ indicates functional uniqueness.

Intuition. When $\alpha > 0$, RASP penalizes the removal of functionally unique structures (low $F(h)$, hence $(1-F(h))^\alpha$ is large, boosting $S(h)$) and facilitates removal of redundant ones. The hypothesis is that recovery (SFT or RLVR) can redistribute the function of pruned redundant structures but cannot recreate unique ones.

3.2 Baselines

- **Standard (Wanda)**: $S(h) = I(h)$, i.e., $\alpha = 0$. Importance-only pruning.
- **GRASPrune**: Gradient-based budgeted structured pruning (ACL 2026 SOTA).
- **Bonsai**: Gradient-free structured pruning (ICML 2025).
- **Unstructured**: Magnitude-based unstructured pruning (accuracy upper bound without speedup).
- **Random**: Uniform random structure removal (lower bound).

4 Experimental Setup

Model. DeepSeek-R1-Distill-Qwen-7B, a widely-used open-source reasoning LLM.

Pruning. Structured pruning of MLP channel groups at 20%, 25%, and 30% sparsity. At these sparsity levels, all 200 pruned structures are MLP groups (0/112 attention GQA groups pruned under any method), as attention groups consistently rank higher in the global importance list. Calibration set: 512 samples from the training distribution.

Recovery. Two paradigms: (1) **SFT**: supervised fine-tuning on GSM8K training solutions for 1870 steps; (2) **SFT→GRPO**: SFT warmup followed by GRPO reinforcement learning with verifiable reward. Both use identical data budgets.

Evaluation. GSM8K (N=1319, full test set) as primary metric. Out-of-distribution (OOD) evaluation on MATH-500 and HumanEval. All results report 95% bootstrap confidence intervals. Multi-seed experiments (5 seeds: 42, 123, 456, 789, 1024) for statistical robustness.

Statistical Analysis. Paired t-tests and Wilcoxon signed-rank tests for pairwise comparisons. 2×2 ANOVA for pruning×recovery interaction. Mixed-effects regression with layer as random effect for per-structure CKA-recoverability analysis.

Model	Accuracy	95% CI	vs Dense
Dense Baseline	76.4%	±2.3pp	—
RASP-20	29.3%	±2.5pp	−47.1pp
Standard-20	28.1%	±2.5pp	−48.3pp
Δ (RASP−Std)	+1.2pp		p=0.34

Table 1: Post-prune GSM8K accuracy (N=1319) at 20% structured sparsity. The RASP–Standard gap is not statistically significant.

Seed	RASP-20	Standard-20	Δ
42	29.3%	28.1%	+1.2pp
123	28.5%	27.5%	+1.0pp
456	30.1%	29.2%	+0.9pp
789	27.8%	26.9%	+0.9pp
1024	28.9%	28.3%	+0.6pp
Mean±Std	28.9±0.8%	28.0±0.8%	+0.9±0.6pp
Paired t-test: p=0.15; Wilcoxon: p=0.19; Cohen's d=0.35			

Table 2: Multi-seed (N=5) post-prune GSM8K accuracy. The RASP advantage is not statistically significant. The seed-to-seed variance (±0.8pp) is comparable to the gap itself (0.9pp).

5 Results

5.1 Post-Prune Accuracy: RASP vs. Standard

Table 1 shows post-prune GSM8K accuracy at 20% sparsity. The gap between RASP-20 and Standard-20 is small (1.2pp) and statistically non-significant (p=0.34, bootstrap).

Multi-seed evaluation (Table 2) confirms this null result across 5 random seeds: the mean gap is 0.9±0.6pp with a paired t-test p=0.15, Wilcoxon p=0.19, and Cohen's d=0.35 (small effect).

5.2 SFT Recovery

SFT recovery is highly effective for both pruning methods, but Standard recovers *better* than RASP (Table 3). Standard-20 reaches 63.5% after SFT (+35.4pp recovery gain), while RASP-20 reaches only 55.3% (+26.0pp). Both models achieve identical training loss (0.64) and token accuracy (83%), indicating the performance gap is not due to optimization differences but to the representational properties of the pruned structures.

5.3 GRPO Recovery

Adding GRPO after SFT provides minimal benefit (Table 4). RASP gains +2.5pp (55.3%→57.8%), but Standard slightly regresses by −1.4pp (63.5%→62.1%). GRPO does not reverse the ranking established during SFT.

5.4 Full Pipeline Comparison

Table 5 presents the complete pipeline comparison. The best overall model is **Standard-20 + SFT** (63.5%), the simplest pipeline. Recovery rate is defined as $(\text{post-recovery} - \text{post-prune})/(\text{dense} - \text{post-prune})$.

Model	Post-Prune	Post-SFT	Recovery Gain
RASP-20	29.3%	55.3%	+26.0pp
Standard-20	28.1%	63.5%	+35.4pp

Table 3: SFT recovery results (N=1319). Standard recovers 8.2pp more than RASP despite similar post-prune accuracy.

Model	Post-SFT	Post-GRPO	GRPO Gain
RASP-20	55.3%	57.8%	+2.5pp
Standard-20	63.5%	62.1%	−1.4pp

Table 4: GRPO stage-2 recovery (N=1319). GRPO provides marginal or negative benefit.

5.5 Pruning×Recovery Interaction

A 2×2 factorial analysis (Table 6) reveals a significant main effect of pruning method (Standard > RASP, p=0.03), no main effect of recovery method (SFT ≈ SFT→GRPO, p=0.62), and no interaction (p=0.41). This directly refutes the hypothesis that the optimal pruning strategy depends on the recovery method.

5.6 Additional Pruning Baselines

We evaluate GRASPrune (ACL 2026 SOTA) and Bonsai (ICML 2025) through the full pipeline (Table 7). GRASPrune exhibits the most severe post-prune collapse (7.2%) yet recovers to 56.0% after SFT—a +48.8pp gain that dramatically demonstrates SFT's power. Bonsai follows a similar pattern. GRPO provides no additional benefit for either method.

The weak correlation between post-prune and post-recovery accuracy across methods is notable: GRASPrune at 7.2% post-prune recovers to within 7.5pp of Standard at 28.1% post-prune after SFT. Post-prune accuracy is a poor proxy for recovery potential.

5.7 Scoring Ablation and α Sensitivity

Scoring ablation. Table 8 compares scoring methods at 20% sparsity. Random pruning (14%) confirms that pruning order matters. Magnitude-only (27%), gradient-only (26%), and Standard/Wanda (28%) perform similarly. RASP with α=4 achieves 40%, a +12pp improvement that constitutes the study's primary positive finding.

α sensitivity. Table 9 shows the full α sweep. Post-prune accuracy increases monotonically with α, but post-SFT accuracy is *non-monotonic*: it peaks at α=0 (Standard, 64%) and degrades for higher α values. This decoupling—optimizing post-prune does not optimize post-recovery—is a central finding.

The implication is that α-scaling reshapes the importance score distribution—aggressively protecting high-importance structures—but the structures it over-protects may be less receptive to SFT gradient signals than those it removes.

5.8 CKA–Recoverability Correlation

The central theoretical claim of recovery-aware pruning is that functional overlap predicts recoverability. We test this via mixed-effects regression on all 784 MLP groups with layer as random effect (Table 10).

CKA overlap $F(h)$ has a non-significant coefficient (β=0.08, p=0.12) and contributes only 1.8% incremental R^2 beyond traditional importance metrics. Gradient importance alone explains 14% of variance. The Spearman correlation between CKA overlap and recoverability is ρ=0.11 (p=0.09)—borderline but not significant. Removing CKA from the model barely changes R^2 (0.34→0.32).

Pipeline	Prune	Recover	Rec. Rate
Dense (no pruning)	—	76.4%	—
RASP + SFT	29.3%	55.3%	55.4%
RASP + SFT→GRPO	29.3%	57.8%	60.7%
Standard + SFT	28.1%	63.5%	73.0%
Standard + SFT→GRPO	28.1%	62.1%	70.1%

Table 5: Full pipeline comparison (N=1319, 20% sparsity). Standard + SFT achieves the highest post-recovery accuracy and recovery rate.

	SFT-Only	SFT→GRPO	Δ
RASP-20	55.3%	57.8%	+2.5pp
Standard-20	63.5%	62.1%	−1.4pp
Δ(Std−RASP)	+8.2pp	+4.3pp	
ANOVA Results:			
Pruning method: F=5.12, p=0.03, η^2=0.08			
Recovery method: F=0.24, p=0.62, η^2=0.004			
Interaction: F=0.68, p=0.41, η^2=0.02			

Table 6: 2×2 factorial analysis (N=1319). Pruning method matters; recovery method does not; they do not interact.

CKA reliability. This null result is not due to measurement noise. CKA scoring is highly reliable: split-half Spearman $\rho = 0.93$ ($p<0.0001$), calibration size sensitivity is low (256 vs. 1024 rank ρ=0.94), and the coefficient of variation across splits is only 0.04. The CKA *measurement* is reliable; the measured property simply does not predict what we need it to predict.

5.9 Sparsity Scaling

We test whether our findings hold at higher sparsity levels (Table 11).

At 25% sparsity, the same pattern holds: RASP has a small post-prune advantage (+1.7pp, not significant) that reverses after SFT recovery (Standard leads by 4pp). GRPO again provides ∼1pp marginal gain. At 30%, both methods collapse to near-random (4–5%), with EOS rates dropping to 8–15%, indicating severe generation impairment.

The sparsity response curve reveals a phase transition: 20% (∼29%) → 25% (∼17%) → 30% (∼5%) → 35% (∼0%). The sparsity ratio is the dominant factor in both post-prune and post-recovery outcomes, dwarfing the effect of pruning method selection.

5.10 Out-of-Distribution Evaluation

To rule out GSM8K-specific memorization, we evaluate recovered models on MATH-500 and HumanEval (Table 12).

Standard's advantage over RASP is consistent across domains: +5pp on MATH-500 and +6pp on HumanEval (SFT recovery). This confirms that Standard's SFT recovery builds genuinely better reasoning ability, not GSM8K-specific pattern matching. All recovered models suffer substantial OOD degradation relative to the dense baseline (e.g., MATH-500: 25–30% vs. 52%), indicating that pruning-then-recovery fragility under distribution shift is a concern regardless of pruning method.

5.11 Unstructured Pruning Baseline

Unstructured magnitude pruning at 20% retains far more post-prune accuracy (54.5%) than any structured method, as it removes individual weights rather than entire groups (Table 13). After SFT+GRPO

Method	Post-Prune	Post-SFT	Post-GRPO
Standard-20	28.1%	63.5%	62.1%
RASP-20	29.3%	55.3%	57.8%
GRASPrune-20	7.2%	56.0%	55.0%
Bonsai-20	25.0%	59.0%	58.0%

Table 7: Four pruning methods through the full pipeline (20% sparsity). Post-prune accuracy poorly predicts post-recovery accuracy: GRASPrune has the worst post-prune (7.2%) but competitive recovery (56%).

Scoring Method	Post-Prune	vs Standard
Random	14%	−14pp
Gradient-only	26%	−2pp
Magnitude-only	27%	−1pp
Standard (Wanda)	28%	—
RASP α=1.0	30%	+2pp
RASP α=4.0	40%	+12pp

Table 8: Scoring ablation (N=100, 20% sparsity). α-scaling substantially improves post-prune accuracy.

recovery, unstructured reaches 63.5%—matching Standard structured + SFT. However, unstructured pruning yields negligible speedup (1.02×) and no memory savings, as sparse weight matrices do not map efficiently to GPU hardware.

This comparison highlights the power of SFT recovery: structured pruning incurs a 27pp post-prune deficit relative to unstructured (28% vs. 55%), yet SFT bridges this gap entirely for Standard pruning (both reach 63.5%).

5.12 Inference Efficiency

Table 14 shows wall-clock inference benchmarks. Both structured pruning methods provide identical speedup (∼1.13–1.14×) and memory reduction (∼18.5%), as they prune the same number of structures with different selection criteria. The pruning method choice does not affect efficiency.

6 Discussion

Why does CKA overlap fail to predict recoverability? CKA measures inter-head functional similarity within a layer, capturing local representational redundancy. However, SFT and GRPO recovery operate through global optimization of the loss landscape. A structure's local functional uniqueness does not determine how readily the *entire network* can compensate for its removal during fine-tuning. The significant layer random effect variance (0.15) in our regression suggests that per-layer properties matter more than per-structure CKA scores. Recovery appears driven by the training signal (data quality and optimization trajectory) rather than the fine-grained structural composition of the pruned model.

Why does Standard outperform RASP after recovery? RASP preferentially protects functionally unique structures (low $F(h)$). But functional uniqueness may correlate with structures that encode specialized, task-specific computations that are *less responsive* to SFT gradient signals. Standard pruning, by contrast, may retain structures that—while functionally redundant—are more amenable to SFT-driven reorganization. The identical training loss and token accuracy (0.64, 83%) for both methods confirm the gap is representational, not optimization-related.

α	Post-Prune	Post-SFT	Post-GRPO
0 (Standard)	28%	64%	63%
0.25	28%	63%	62%
0.5	29%	62%	62%
1.0	30%	59%	60%
2.0	34%	60%	60%
4.0	40%	62%	59%

Table 9: α ablation. Post-prune accuracy monotonically increases with α, but post-recovery accuracy does not.

Predictor	β	SE	p	Incr. R^2
Gradient Importance	0.31	0.06	<0.001	0.14
Weight Norm	0.23	0.05	0.001	0.09
Activation Magnitude	0.19	0.05	0.004	0.06
CKA Overlap $F(h)$	0.08	0.05	0.12	0.018
Full model R^2				0.34
Model w/o CKA R^2				0.32

Table 10: Mixed-effects regression predicting per-structure post-recovery accuracy retention (N=784 MLP groups). CKA functional overlap is not a significant predictor.

Why does α-scaling help post-prune but not post-recovery? High α aggressively concentrates importance scores, protecting high-importance structures at the expense of marginal ones. This preserves more immediate reasoning capacity (post-prune), but the over-protected structures may not be the ones most critical for SFT-mediated recovery. The monotonic post-prune improvement with decoupled post-recovery behavior suggests that post-prune accuracy is optimizing a proxy objective that diverges from the true target.

Positive takeaways. Despite the negative headline findings, this study offers actionable conclusions: (1) α-scaled importance scoring is a simple, effective way to improve post-prune accuracy when recovery is not planned; (2) Standard Wanda pruning + SFT is a strong default pipeline that practitioners should prefer over more complex alternatives; (3) GRPO adds negligible value as a recovery method at moderate sparsity, saving substantial compute; (4) sparsity ratio is the dominant lever—practitioners should focus on choosing the right compression target rather than optimizing pruning criteria.

Limitations. Our study uses a single base model (DeepSeek-R1-Distill-Qwen-7B) and tests sparsity levels up to 30%. Higher sparsity ($\geq$40%) may enter a regime where attention groups are pruned and CKA signals become relevant. The SFT and GRPO configurations use fixed hyperparameters; alternative recovery protocols might yield different interactions. The study focuses on math reasoning; generalization to other reasoning domains (code, logic) requires further investigation.

7 Conclusion

We conducted a comprehensive empirical study testing whether reasoning LLMs benefit from recovery-aware structured pruning. Across 5 seeds, 3 sparsity levels, 4 pruning methods, and 2 recovery paradigms, we find that CKA-based functional overlap does not significantly predict recoverability, and the simplest pipeline—Standard importance pruning with SFT recovery—achieves the best results. GRPO recovery provides no additional benefit over SFT. While α-scaled importance scoring effectively improves post-prune accuracy, this advantage does not survive recovery. These negative results challenge the intuitive

Model	Post-Prune	Post-SFT	Post-GRPO
25% Sparsity			
RASP-25	17.2%	48%	49%
Standard-25	15.5%	52%	53%
Δ (RASP$-$Std)	+1.7pp	$-$4pp	$-$4pp
30% Sparsity (probe, no recovery)			
RASP-30	5%	—	—
Standard-30	4%	—	—

Table 11: Higher sparsity results. The pattern from 20% replicates at 25%: Standard recovers better. At 30%, both methods near-collapse.

Model	GSM8K	MATH-500	HumanEval
Dense	76.4%	52%	58%
Standard + SFT	63.5%	30%	37%
Standard + GRPO	62.1%	28%	35%
RASP + SFT	55.3%	25%	31%
RASP + GRPO	57.8%	24%	31%

Table 12: OOD evaluation. Standard's advantage is consistent across all benchmarks, ruling out memorization.

appeal of recovery-aware pruning and suggest that, at moderate sparsity, SFT recovery is robust enough to dominate pruning structure selection as the primary determinant of final model quality.

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Method	Post-Prune	Post-Recov.	Speedup
Unstructured 20%	54.5%	63.5%	1.02×
RASP-20 (structured)	29.3%	57.8%	1.13×
Standard-20 (structured)	28.1%	63.5%	1.14×
GRASPrune-20 (struct.)	7.2%	55.0%	1.13×

Table 13: Unstructured vs. structured pruning. Structured pruning + SFT recovery matches unstructured accuracy while providing real speedup.

Model	Latency (ms/tok)	Throughput (tok/s)	Memory (GB)
Dense	28.5	35.1	14.2
RASP-20	25.2	39.7	11.6
Standard-20	25.0	40.0	11.5

Table 14: Inference benchmarks. Both pruning methods yield identical speedup ($\sim$1.13×) and $\sim$18.5% memory reduction.

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